

Umrigar Kakshil

Final Year, RGIPT | Chemical Engineering Undergraduate

Male | 22-01-2005

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Objective

"Analytical Chemical Engineering Undergraduate with practical exposure to petrochemical manufacturing seeking to leverage structured problem-solving skills and process optimization experience towards global energy transition pathways."

Education

Examination	Institution	Year	CPI / %
Graduation	Rajiv Gandhi Institute of Petroleum Technology	2023-27	6.66 (till 6 th sem)
Intermediate	Pioneer Convent School, Indore	2022	83.4
Matriculation	Reliance Foundation School, Surat	2020	90.8

Internships

GET Summer Intern, Reliance Industries Ltd.

(May26'-July26')

Vadodara Manufacturing Division, Vadodara | Mentor: Mr. Ramesh Jethva

- Spearheaded a techno-economic evaluation of the cycle water loop at a 23,300 MTA EO/EG plant, analyzing the transition from an atmospheric cooling tower to a closed-loop gasketed Plate and Frame Heat Exchanger.
- Modeled process heat duties and mass balances, simulated a 225 m² Plate and Frame Heat Exchanger on HTRI, and analyzed operational cost savings (blowdown reduction, eliminated fan power) to project a 4-year payback.

Vocational Trainee, Reliance Industries Ltd.

(May25'-July25')

Hazira Manufacturing Division, Surat | Mentor: Mr. Tapan P. Mankad

- Analyzed process operations and operational drivers at RIL's Hazira Cracker Unit, focusing on feedstock-to-product flows, energy efficiency, and safety compliance.
- Monitored process parameters for 10+ cracking furnaces, 5+ fractionators, and 2+ compressors to bridge theoretical chemical engineering concepts with large-scale industrial execution.

Projects

Green Hydrogen Project

Guide: Dr. M. S. Balathanigaimani

(February26' - Present)

- Designing a comparative techno-economic framework to evaluate industrial-scale Green Hydrogen production pathways and electrolyzer technologies for hard-to-abate refinery applications.
- Synthesizing policy frameworks of India (SIGHT program), Australia, and Japan to analyze commercial viability, capital subsidies, and cross-border supply chain integration.
- Modeling the impact of electricity tariffs, electrolyzer CAPEX, and capacity factors on the Levelized Cost of Hydrogen (LCOH) to identify regional cost-reduction pathways.

Skills

- **Technical Skills:** HTRI, MS Office (Excel, PowerPoint, Word), Aspen Plus, Generative AI, Prompt Engineering, AutoCAD.
- **Methodologies:** Techno-Economic Evaluation, Comparative Analysis, Literature Synthesis & Industry Research
- **Soft Skills:** Adaptability, Teamwork, Time Management, Leadership, Problem Solving, Project Handling.

Achievements and Certificates

- Completed "Google AI Essentials" and "Google Prompting Essentials" certifications by Google.
- Qualified prestigious exams like JEE, BITSAT, IAT. (2023)
- Qualified prestigious Olympiads like NTSE Stage I, NSO*, IMO*. (2019)

Extra-Curricular Activities/PORs

Editorial Executive – IChE RGIPT Student Chapter

(June 2024 – June 2025)

Logistics and Hospitality Executive – SCHEMCON 2024

(Feb 2024 - Sep 2024)